

Ram Chandra Beniwal

AI/ML Lead | LLM & Agentic Systems Engineer | RAG & Multi-Agent Systems | Senior Unity3D/XR Developer

ramchandra.nitsri@gmail.com | +91-8290637199 | LinkedIn

Professional Summary

AI/ML Lead with 4+ years of experience building and shipping **production LLM systems: multimodal RAG pipelines, agentic orchestration, multi-agent systems, and knowledge-graph-powered retrieval**. Currently leading a full AI engineering team at Futureverse, owning architecture from ingestion and hybrid retrieval (BM25 + vector) through reranking, evaluation, and deployment. Deep hands-on expertise in **RAG, GraphRAG, AI Agents, Knowledge Graphs, embeddings, and RAG evaluation** using OpenAI API, LangChain, Weaviate, Neo4j, and Hugging Face. Unique background bridging intelligent AI backends with real-time 3D/XR frontends for medical training and surgical visualization products.

Core Technical Skills

- **LLM & Agentic Systems:** RAG, Multimodal RAG, GraphRAG, Agentic Orchestration, Multi-Agent Systems, AI Agents, Tool Use / Function Calling, Prompt Engineering
- **Retrieval & Search:** Hybrid BM25 + Vector Search, Semantic Retrieval, Multi-Query Expansion, Cross-Encoder Reranking, Relevance Thresholding, Chunking Strategies, Embedding Pipelines (bge-m3)
- **Knowledge Graphs:** Neo4j, AutoSchema Extraction, Hybrid Vector + Graph Search
- **AI Frameworks & APIs:** LangChain, PyTorch, Hugging Face Transformers, OpenAI API, Vision LLMs (image captioning/understanding)
- **Evaluation & Reliability:** RAG Eval Suites (faithfulness, completeness, citation accuracy), Hallucination Mitigation, Production Debugging
- **Databases:** Weaviate, Pinecone, FAISS, Neo4j, SQL/MySQL
- **Programming:** Python, C# (Advanced), C++, JavaScript (ES6+), SQL
- **3D/XR:** Unity3D (4+ years), OpenXR, MRTK, XR Interaction Toolkit, ARCore, WebGL/Three.js
- **Cloud & DevOps:** Docker, AWS (EC2, S3, Lambda), CI/CD, Git, Linux, REST APIs

Professional Experience

AI/ML Lead | Futureverse Private Limited

Feb 2026 – Present

- Lead the AI engineering team end-to-end: architecture decisions, sprint planning, code reviews, and mentoring across RAG, agentic orchestration, and multi-agent system initiatives.
- Designed **agentic orchestration and multi-agent workflows** coordinating specialized LLM agents for retrieval, reasoning, and response generation in production.
- Built a **multimodal RAG pipeline** — heading-based chunking with multi-granularity metadata, hybrid BM25 + vector indexing (Weaviate, bge-m3), and vision-LLM image captioning embedded in the same vector space as text for unified ranked retrieval.
- Fused surrounding document context into image embeddings and added LLM-based decorative-image filtering, overcoming vision-model recognition limits and improving retrieval precision.
- Added multi-query expansion with cross-encoder reranking and relevance-threshold cutoffs to eliminate off-topic and hallucinated citations.
- Auto-rendered inline citations and images in the UI, backed by a **RAG eval suite** scoring faithfulness, completeness, and citation accuracy.
- Fixed a critical vector-DB schema bug causing silent retrieval failures, restoring production search reliability.

Team Lead – Unity, XR & AI/LLM Development | Metamix Technologies

Jun 2023 – Feb 2026

- Led Unity + AI development for medical XR products: **AnatomyXR, Medvisor, BirthXR, LivXR**.
- Built **RAG and GraphRAG pipelines** (OpenAI API, LangChain, Neo4j) powering AI-driven medical knowledge retrieval and in-app anatomy Q&A inside XR applications.
- Developed an **LLM-based question and quiz generator** for AnatomyXR, auto-creating assessment content grounded in the medical knowledge base via retrieval-augmented prompting.
- Implemented **AutoSchema Extraction** workflows to automatically generate Neo4j graph schemas from unstructured medical content.
- Designed modular Unity architecture using design patterns; built client-server multi-user XR applications.

- Optimized applications for Windows desktop, mobile, and XR devices; integrated backend SQL data systems.
- Collaborated with doctors, researchers, 3D modelers, and UI/UX teams across the full software development lifecycle.

Unity Game & Web Developer | Parcellinc

Jan 2022 – Jun 2023

- Developed Unity-based games (**Zombie Runner**, **Realm Rush**) with AI-driven gameplay mechanics.
- Implemented AI behavior trees and decision systems for enemy NPC logic and adaptive difficulty.
- Built WebGL and Three.js interactive 3D web experiences; developed cross-platform Android applications.
- Improved performance, resolved bugs, and optimized rendering pipelines in agile cycles with designers and 3D artists.

Key Projects

Multimodal Enterprise RAG Platform – Futureverse

- Unified text and image search in a single vector space with hybrid BM25 + vector retrieval, cross-encoder reranking, inline cited answers, and a full evaluation suite; orchestrated via a multi-agent pipeline.

AnatomyXR – Immersive Anatomy Learning with AI Tutor (Meta Quest)

- Built layered organ interaction and gesture-based XR modules; optimized high-poly anatomical assets.
- Integrated **RAG pipeline with Neo4j Knowledge Graph** for AI-driven anatomy Q&A and an **automatic quiz/question generator** inside the XR environment.

Medical Knowledge Graph System – AutoSchema & Semantic Retrieval

- Designed **AutoSchema Extraction pipeline** to parse and structure unstructured clinical documents into Neo4j graphs, enabling hybrid vector + graph retrieval.

Medvisor – Mixed Reality Surgical AI Assistance

- Developed MR surgical visualization and contextual simulation workflows validated with medical professionals.

Education

B.Tech – Chemical Engineering

National Institute of Technology, Srinagar

2014 – 2018

CGPA: 7.41